

BHUMIBOL DAM, ThailandWMO: **483770**Lat: **17.2439N**Lon: **99.0025E**Elev: **474**StdP: **14.45**Time Zone: **7.00 (E07)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	58.6	61.4	49.5	53.3	82.0	52.4	59.4	79.8	7.7	82.4	7.1	81.9	0.0	310	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.4	102.7	76.5	100.8	76.4	98.7	76.3	83.1	94.0	81.5	92.1	80.3	90.8	2.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.2	159.6	90.6	78.8	152.4	87.3	77.8	147.3	85.3	47.6	93.9	45.7	92.6	44.4	90.7	89.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	8.7	7.3	DB	54.4	105.5	3.0	2.3	52.3	107.2	50.6	108.5	48.9	109.8	46.7	111.5	(4)
(5)			WB	51.9	84.3	3.7	2.2	49.3	85.8	47.1	87.1	45.1	88.3	42.4	89.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	76.5	80.7	85.9	88.8	86.5	84.5	83.7	83.5	83.0	81.3	78.8	75.5	(6)
(7)		DBStd	5.20	3.85	3.96	4.17	4.21	4.53	3.07	2.83	2.51	2.38	2.38	3.20	3.95	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	2	1	0	0	0	0	0	0	0	0	0	0	1	(9)
(10)		CDD50	11821	821	860	1113	1163	1132	1034	1045	1038	990	970	864	791	(10)
(11)		CDD65	6347	357	440	648	713	667	584	580	573	540	505	414	326	(11)
(12)		CDH74	71079	2968	4830	8742	10038	8458	6936	6636	6397	5655	4554	3384	2481	(12)
(13)	CDH80	34201	1263	2524	5130	6114	4544	3065	2725	2540	2242	1758	1363	933	(13)	
(14)	Wind	WSAvg	1.2	0.9	1.3	1.7	1.7	1.5	1.7	1.9	1.8	1.0	0.4	0.4	0.5	(14)
(15)	Precipitation	PrecAvg	42.7	0.3	0.3	1.1	2.5	7.2	4.6	3.8	4.8	9.1	7.2	1.3	0.3	(15)
(16)		PrecMax	60.5	2.4	3.5	7.6	7.1	17.5	8.9	8.2	9.3	15.9	12.3	5.9	2.3	(16)
(17)		PrecMin	29.3	0.0	0.0	0.0	0.2	1.3	0.9	1.4	1.9	2.5	1.0	0.0	0.0	(17)
(18)		PrecStd	8.7	0.6	0.7	1.8	1.9	4.6	2.4	1.9	1.8	3.4	3.6	1.4	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.4	99.7	103.5	106.1	104.6	98.7	96.6	96.2	95.3	93.9	93.9	92.0	(19)
(20)			MCWB	70.7	71.8	73.7	77.0	78.9	79.3	79.2	79.0	79.4	79.5	77.7	74.4	(20)
(21)		2%	DB	92.0	97.1	101.6	103.6	101.1	95.7	93.9	93.7	93.3	92.0	91.6	89.9	(21)
(22)			MCWB	71.2	71.5	73.6	77.3	79.1	79.0	78.6	78.5	79.1	78.9	77.3	74.0	(22)
(23)		5%	DB	89.8	95.1	99.7	101.5	98.4	93.3	91.9	91.6	91.6	90.4	89.7	87.9	(23)
(24)			MCWB	71.3	71.7	73.6	77.5	79.3	78.7	78.0	78.2	78.8	78.5	76.4	73.0	(24)
(25)		10%	DB	87.5	92.7	97.6	99.2	95.6	91.1	89.8	89.5	89.7	88.6	87.6	85.6	(25)
(26)			MCWB	71.0	71.4	73.6	77.5	78.9	78.3	77.5	77.6	78.4	78.0	75.8	72.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.8	79.4	82.6	85.1	86.0	82.7	82.5	82.7	82.6	82.4	81.5	79.0	(27)
(28)			MCDWB	87.1	92.6	95.8	99.2	97.4	91.7	91.4	90.7	90.2	90.2	90.5	88.0	(28)
(29)		2%	WB	75.1	76.5	79.4	82.3	82.6	80.9	80.1	80.0	80.7	80.1	79.2	76.6	(29)
(30)			MCDWB	85.5	89.4	93.6	96.3	93.9	90.2	89.4	88.7	89.4	89.0	88.0	85.8	(30)
(31)		5%	WB	73.8	75.0	77.4	80.2	80.6	79.9	79.0	79.1	79.8	79.2	78.0	75.2	(31)
(32)			MCDWB	84.1	88.4	92.4	94.7	92.7	89.1	88.3	87.9	88.3	87.9	86.5	83.6	(32)
(33)		10%	WB	72.5	73.6	76.0	78.8	79.5	79.1	78.1	78.3	79.0	78.5	76.8	73.7	(33)
(34)			MCDWB	83.0	87.0	91.2	93.9	91.0	88.1	87.1	86.7	87.3	86.7	84.6	82.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.8	24.7	22.9	20.4	15.9	12.7	11.9	12.3	13.6	14.6	17.6	20.5	(35)
(36)			MCDBR	25.2	26.6	25.2	22.4	19.6	16.0	14.9	15.5	16.4	17.3	19.8	22.1	(36)
(37)			MCWBR	9.0	8.7	7.5	6.5	5.8	5.2	5.2	5.1	5.6	6.2	7.5	8.2	(37)
(38)		5% WB	MCDBR	21.2	23.5	22.4	21.3	17.6	14.0	13.5	13.8	14.8	15.8	17.4	18.8	(38)
(39)			MCWBR	9.2	10.0	9.0	8.4	6.9	5.6	5.5	5.4	5.9	6.4	7.7	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.467	0.559	0.683	0.642	0.504	0.444	0.433	0.440	0.474	0.544	0.469	0.464	(40)	
(41)		taud	2.126	1.896	1.637	1.778	2.185	2.382	2.412	2.379	2.269	2.018	2.203	2.157	(41)	
(42)		Ebn at Noon	258	240	215	226	256	269	272	273	264	240	256	255	(42)	
(43)		Edh at Noon	47	62	83	72	48	39	38	39	43	54	43	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1489	1689	1768	1900	1688	1440	1339	1330	1428	1453	1495	1421	(44)	
(45)		RadStd	71	68	111	109	154	134	123	95	80	86	93	68	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	+0.20	N/A	N/A	+218	+218

Nomenclature: See separate page